

Fetal Health Classification from Cardiotocogram Data

1. Problem Statement

A cardiotocogram (CTG) records the fetal heart rate and uterine contractions during pregnancy. Reading it by hand is slow and subjective. The task is to classify each CTG record as **Normal**, **Suspect** or **Pathological** from its numeric features. We reimplement the reference paper and check whether our accuracies match the ones it reports.

2. Dataset

UCI Cardiotocography data (*fetal_health.csv*): 2,126 records and 21 features, each labelled by expert obstetricians. After preprocessing, 2,113 records were used and split 70/30 into 1,479 training and 634 test samples. The classes are imbalanced (about 78% Normal):

	Normal	Suspect	Pathological	Total
Train	1,149	202	128	1,479
Test	497	90	47	634

Features were reduced to 10 with SelectKBest and standardised with StandardScaler.

3. Approach

We trained ten classifiers on the same train/test split, using library defaults where the paper does not give hyperparameters. Each model is scored on test accuracy, with per-class precision, recall and F1 and confusion matrices as supporting views. Accuracies are then compared with the values reported in the paper. After this baseline we tried to improve on it with hyperparameter search, class weights, SMOTE, voting and stacking, and a teammate ran a soft-voting ensemble on all 21 features.

4. Implementation

Python in Google Colab with scikit-learn, XGBoost and LightGBM. Preprocessing and feature selection write processed train and test CSV files, which the two model notebooks read. Notebook 03 trains an MLP (one hidden layer of 100 units), Gradient Boosting, XGBoost, LightGBM and a Linear SVM. Notebook 04 trains Logistic Regression, KNN, Decision Tree, Random Forest and an RBF SVM. Both notebooks end with a comparison against the paper. A third notebook holds the tuning and ensemble experiments, and a separate one the soft-voting run.

5. Results (test accuracy, %)

Model	Ours	Paper	Difference (pp)
LightGBM	95.90	95.58	+0.32
Gradient Boosting	95.58	95.58	+0.00
XGBoost	95.58	94.95	+0.63
Random Forest	95.27	94.95	+0.32
MLP	94.95	93.53	+1.42
KNN	93.85	94.01	-0.16
Decision Tree	93.22	93.38	-0.16
SVM (RBF)	91.96	92.11	-0.15
Logistic Regression	89.59	89.59	+0.00
Linear SVM	89.27	89.27	+0.00

6. Further Experiments

Baselines for comparison: KNN 93.85, SVM 91.96, Random Forest 95.27, best single model LightGBM 95.90. The soft-voting run averages the class probabilities of three boosted models with equal weights on all 21 features, with no scaling, and uses a new stratified 70/30 split (seed 42) which gave an accuracy of 96%.

Experiment (10 features unless noted)	Accuracy (%)
KNN, grid search (k=3, distance, Manhattan)	94.01
SVM (RBF), grid search (C=100)	95.11
Random Forest, 300 trees, fixed seed	95.58
Random Forest, balanced class weights	95.27
Random Forest, SMOTE oversampling	94.95

Random Forest, all 21 features	94.95
Random Forest, random search (CV)	95.11
Soft voting: LR + KNN + RF + SVM	95.11
Stacking: same four, LR meta-model	95.27
Soft voting: LightGBM + XGBoost + CatBoost, all 21 features (teammate)	96

7. Conclusions

- Boosted trees and Random Forest did best: LightGBM reached 95.90% (608 of 634 correct). The linear models stayed near 89%. Nine of the ten models are within 0.65 points of the paper; the MLP differs most (+1.42), as the paper gives no network size or seed.
- Tuning helped the weakest models most (SVM +3.2 points). Class weights, SMOTE, voting and stacking of classic models all landed between 94.9 and 95.6, no better than a seeded Random Forest.
- The soft-voting ensemble of LightGBM, XGBoost and CatBoost on all 21 features reached 96%, the best result overall. The gain over single LightGBM is only a few test samples (one sample is 0.16 points) and the split differs, so it should not be over-read.
- Suspect is the hardest class: LightGBM recalls only 78.9% of Suspect cases, against 99.0% for Normal.
- Limitations: one split, no cross-validation for the final scores. Some choices (such as the Random Forest seed and tree count) were compared on the test set, so those numbers are slightly optimistic. Unseeded models vary by about 0.6 points between runs.